`Appendix`: Federated Foundation Models on Heterogeneous Time Series

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This appendix contains additions to the AAAI 2025 submission "Federated Foundation Models on Heterogeneous Time Series", including:

A. More Related Work
B. Technical and Implementation Details
C. Theorem and Proofs
D. Full Experiment Results

- Reproducibility Checklist
- Statement for Reproducibility Checklist

A. More Related Work

Time Series Foundation Models Pre-trained models with scalability can evolve into foundation models, characterized by increasing model capacity and pre-training scale, enabling them to address diverse data. Large Language Models (LLMs) have demonstrated advanced capabilities, including in-context learning and few/zero-shot learning (Jin et al. 2023). However, due to the heterogeneity of time series data, the development of GPT-style time series foundation models (TSFMs) has been limited. Research on TSFMs is in its early stages, with existing efforts categorized into two groups, one of which is LLM-empowered time series analysis. For instance, FPT (Zhou et al. 2023) utilizes GPT-2 as a representation extractor for time series, fine-tuning it on downstream datasets and tasks. LLM4TS (Chang, Peng, and Chen 2023) encodes time series into numerical tokens for LLMs, demonstrating model scalability in forecasting tasks. Time-LLM (Jin et al. 2023) explores prompting techniques to enhance prediction, inspiring cross-modality representation ability through language templates. UniTime (Liu et al. 2024a) introduces cross-domain techniques to fine-tune pre-trained LLMs, reducing prediction biases. Nevertheless, the performance of these LLM-based approaches is heavily dependent on the LLM backbone and cross-modal design. In contrast, our proposed model is pre-trained natively on time series, eliminating the need for extra modality alignment.

Another category of models involves pre-training on large-scale time series datasets. For instance, ForecastFPN (Dooley et al. 2024) is trained on synthetic series to achieve zero-shot forecasting. CloudOps (Woo et al. 2023) employs masked modeling on Transformers for domain-specific forecasting. TimeGPT-1 (Garza and Mergenthaler-Canseco 2023) introduces the first commercial API for zero-shot forecasting, marking a milestone in the field. PreDcT (Das et al. 2023), a decoder-only Transformer pre-trained on Google Trends, demonstrates notable zero-shot capabilities. Furthermore, MOMENT (Goswami et al. 2024) and Timer (Liu et al. 2024b) pre-train on ultra-large time series datasets and release corresponding datasets, showcasing excellent cross-task performance empirically. This suggests that training TSFMs is a resource-intensive endeavor.

B. Technical and Implementation Details

Model Architecture Within our FFTS, we use the standard encoder-only Transformer as the local model, the detailed configuration is shown in **Table 1**.

Component	Configuration
Number of Layer	12
Patch Size	16
Dimension of Model	1024
Dimension of FFN	512
Number of Head (Attention)	8
Dimension of Trend Linear	512
Dimension of Seasonal Linear	512
Dimension of Adaption Head	512
Input Length	512

Table 1: Configuration of the model architecture within our FFTS. The local and global model architectures and configurations are consistent.

Reversible Normalization In time series analysis, statistical properties such as mean and variance often exhibit temporal shifts, indicating distributional changes in the data. To address this, we've incorporated Reversible Normalization (RevIn) (Kim et al. 2021) into our FFTS. Specifically, we've integrated RevIn into local model in each client. This introduces two dynamic factors that adaptively normalize segments of the time series $\boldsymbol{X}$, enhancing the accuracy of potential temporal pattern modeling. Specifically, for the input time series of $\boldsymbol{X}$, its transformed value can be given by:

$$\boldsymbol{X}' = \gamma_T \left(\boldsymbol{X} - \frac{\mathbb{E}[\boldsymbol{X}]}{\sqrt{\text{Var}[\boldsymbol{X}] + \epsilon_T}} \right) + \beta_T \quad (1)$$

where $\mathbb{E}[\boldsymbol{X}]$ and Var $[\boldsymbol{X}]$ are the instance-specific mean and variance, respectively. γ_T and β_T are the trainable parameters for this component.

Time Series Decomposition In proposed gating network in the ATM, we decompose the input sequence into multimodal latent statistical representation to improve the process of achieving the timescale weight,

$$\boldsymbol{X}^k_{\text{Trend}} + \boldsymbol{X}^k_{\text{Seasonal}} = \texttt{Decomp}(\boldsymbol{X}^k), \quad (2)$$

where the $\boldsymbol{X}^k \in \mathbb{R}^{L\times 1}$ denote the k-th variable in time series $\boldsymbol{X} \in \mathbb{R}^{L\times C}$, the trend component $\boldsymbol{X}_{\text{Trend}}$ and the seasonal component $\boldsymbol{X}_{\text{Seasonal}}$ captures the underlying long-term temporal pattern and encapsulates the repeating short-term varibale cycles, respectively. Note that $\boldsymbol{X}_{\text{Trend}}$ and $\boldsymbol{X}_{\text{Seasonal}}$ have the same shape as $\boldsymbol{X}$. This decomposition explicitly enables the identification of unusual observation and shifts in seasonal patterns or trends.

Implementation Details We mainly follow the experimental configurations across all baselines within a unified evaluation pipeline in `Time-Series-Library` for fair comparison. All our experiments are repeat five times and we report the averaged results. All the algorithm implementations and designs in this study are based on Pytorch 1.9.0 and the algorithms are run on a RTX3090 GPUs 24GB. By tradition, our code will be available upon acceptance.

Federated Pretraining The configuration of our FFTS during pre-training is presented in **Table 2**. We selected 18 cross-domain heterogeneous time series datasets for pre-training[1], encompassing various domains such as economy, transportation, health, energy, nature, and web. Each dataset was treated as an independent client, and the details of the pre-training datasets are provided in **Table 3**. During pre-training, each client's dataset was split into training and test subsets with a uniform ratio of 7:3. All experiments were conducted five times with different random seeds, and the mean values of the evaluation metrics were reported.

Downstream Dataset Information The global model obtained from FFTS pre-training, referred to as TSFM, is adapted to classical downstream time series analysis tasks, including forecasting, imputation, and anomaly detection. The datasets used for long-term forecasting are presented in **Table 4**, which are also used for the imputation task, except for the Illness dataset. The datasets for short-period forecasting are listed in **Table 5**. To ensure a fair comparison, our experimental setup for downstream fine-tuning was aligned with that of Time-LLM (Jin et al. 2023) (forecasting and imputation tasks) and GPT4TS (Zhou et al. 2023) (anomaly detection task). The FFTS results reported are the average of the evaluation metrics obtained from five independent runs with different random seeds.

[1]These datasets are publicly available and were sourced from the Monash Time Series Forecasting Repository.

Pretraining Component	Hyper-parameter
Input Length	512
Average Masking Length (L_m)	16
Masking Ratio (r_m)	0.35
Value of k	3
Value of λ	$1e^{-1}$
Learning Rate	$\{1e^{-3}, 5e^{-4}, 1e^{-4}\}$
Batch Size	256
Local Updating Epochs	20
Global Communication Round	50

Table 2: Configuration in the federated pretraining phase.

Baseline We compare two types of baselines in our experiments: federated learning baselines during pretraining and deep time series model baselines in downstream tasks. Details of federated learning baseline are as follows:

- FedAvg (McMahan et al. 2017) A decentralized approach that enables devices to collaboratively learn a shared model by aggregating locally-computed updates, allowing for the training of deep networks on private and large datasets while reducing communication costs.
- FedProx (Li et al. 2020) A federated learning framework that generalizes and re-parameterizes FedAvg to tackle systems and statistical heterogeneity, providing convergence guarantees and demonstrating more robust and accurate convergence behavior.
- pFedMe (T Dinh, Tran, and Nguyen 2020) A personalized federated learning algorithm that uses Moreau envelopes to decouple personalized model optimization from global model learning.

Details of deep time series model baseline is as follows:

- LogTransformer (Li et al. 2019) A modified Transformer architecture that addresses the locality-agnostics and memory bottleneck issues in time series forecasting by incorporating convolutional self-attention and logsparse. attention, achieving improved forecasting accuracy with reduced memory cost.
- N-BEATS (Oreshkin et al. 2020) A deep neural architecture that achieves state-of-the-art performance in univariate time series point forecasting, using a stack of fully-connected layers with backward and forward residual links.
- Reformer (Kitaev, Kaiser, and Levskaya 2020) This model improves Transformer by using locality-sensitive hashing for attention and reversible residual layers. It offers better memory efficiency and speed for lengthy sequences without sacrificing performance.
- Informer (Zhou et al. 2021) An optimized Transformer-based model for long-range time series prediction. It uses ProbSparse self-attention for efficiency, processes long inputs effectively, and employs a fast prediction decoder.
- LightTS (Zhang et al. 2022) A lightweight MLP structure. It utilizes two downsampling strategies—spaced

Domain	Dataset	# Channels	Frequency	Length
Economic	Bitcoin	18	Daily	Min: 2659 Max: 4581
	FRED-MD	107	Monthly	Fixed: 728
	NN5	111	Daily	Fixed: 791
Transport	Pedestrain Counts	66	Hourly	Min: 576 Max: 96424
	Rideshare	2304	Daily	Fixed: 541
	San Francisco Traffic	862	Hourly/Weekly	Fixed: 17544
Health	COVID Deaths	266	Daily	Min: 212 Max: 212
	Hospital	767	Monthly	Fixed: 84
Energy	London Smart Meters	5560	Daily	Min: 288 Max: 39648
	Wind Farms	339	Minutely	Min: 6345 Max: 527040
	Wind Power	1	Second	Fixed: 7397147
	Electricity	321	Hourly/Weekly	Fixed: 26304
Nature	KDD Cup 2018	270	Hourly	Min: 9504 Max: 10920
	Oikolab Weather	3010	Daily	Min: 1332 Max: 65981
	Temperature Rain	32072	Daily	Fixed: 725
Web	Web Traffic	145063	Daily	Fixed: 803

Table 3: Dataset statistics about dataset used in federated pretraining process, each dataset representing a distinct data source stored on a separate client. The channels indicates the number of time series. 'Min' and 'Max' denote the shortest and longest sequence lengths, respectively, while 'Fixed' signifies datasets with uniform sequence lengths.

Dataset	Domain	# Channels	Frequency	Size	Forecast Length
ETTh1	Power	7	Hourly	(8545, 2881, 2881)	$\{96, 192, 336, 720\}$
ETTh2	Power	7	Hourly	(8545, 2881, 2881)	$\{96, 192, 336, 720\}$
ETTm1	Power	7	15 Minute	(34465, 11521, 11521)	$\{96, 192, 336, 720\}$
ETTm2	Power	7	15 Minute	(34465, 11521, 11521)	$\{96, 192, 336, 720\}$
Illness	Epidemiology	7	Weekly	(617, 74, 170)	$\{24, 36, 48, 60\}$
Weather	Weather	21	10 Minute	(36792, 5271, 10540)	$\{96, 192, 336, 720\}$

Table 4: Dataset statistics about long-term forecasting (imputation) dataset. The channels indicates the number of time series (i.e., variables), and the size is organized in (training, validation, testing). Note that Illness dataset is not included in the imputation task.

and sequential sampling—on the MLP structure, capitalizing on the fact that downsampled time series generally maintain most of their original information.

- ETSformer (Woo et al. 2022) A Transformer architecture that leverages the principle of exponential smoothing to improve traditional Transformers for time-series forecasting, offering better decomposition capability, interpretability, and long-term forecasting efficiency.
- Stationary (Liu et al. 2022) A framework that addresses the over-stationarization problem in time series forecasting by combining Series Stationarization and De-stationary Attention modules, which unify statistics for better predictability while recovering intrinsic non-stationary information.
- Autoformer (Wu et al. 2021) A decomposition architecture that leverages leveraging an Auto-Correlation mechanism for long-term forecasting, which efficiently discovers dependencies and aggregates representations at the sub-series level.
- FEDformer (Zhou et al. 2022) A forecasting method that combines seasonal-trend decomposition with a frequency-enhanced Transformer, capturing both the global profile and detailed structures of time series.
- Pyraformer (Liu et al. 2021) It features hierarchical pyramidal attention modules with binary trees to capture temporal dependencies across different ranges efficiently, both in time and memory complexity.
- AnomalyTransformer (Xu et al. 2022) A novel approach for unsupervised time series anomaly detection that leverages the self-attention mechanism to compute association discrepancy, which captures the adjacent-concentration bias of anomalies.
- TimesNet (Wu et al. 2022) Aframework that transforms 1D time series into 2D tensors to model complex temporal variations, leveraging the multi-periodicity of time series to adaptively discover and extract intraperiod- and interperiod-variations.
- Patchtst (Nie et al. 2022) This method divides the time

Dataset	Domain	# Channels	Frequency	Size	Forecast Length
M4-Yearly	Demographic	1	Yearly	(23000, 0, 23000)	6
M4-Quarterly	Finance	1	Quarterly	(24000, 0, 24000)	8
M4-Monthly	Industry	1	Monthly	(48000, 0, 48000)	18
M4-Weekly	Macro	1	Weekly	(359, 0, 359)	13
M4-Daily	Macro	1	Daily	(4227, 0, 4227)	14
M4-Hourly	Other	1	Hourly	(414, 0, 414)	48

Table 5: Dataset statistics about short-term forecasting dataset. The channels indicates the number of time series (i.e., variables), and the size is organized in (training, validation, testing).

series into patches at the sub-series level for input to the Transformer. Each channel holds a univariate time series, sharing the same embedding and Transformer weights across all series.

- DLinear (Zeng et al. 2023) DLinear integrates decomposition schemes from Autoformer and FEDformer with linear layers to model time series data tables. It effectively summarizes trend and seasonal components, enhancing performance on datasets rich in trends.

- N-HiTS (Challu et al. 2022) A forecasting model that addresses the challenges of long-horizon forecasting by incorporating hierarchical interpolation and multi-rate data sampling techniques.

- GPT4TS (Zhou et al. 2023) This model is designed for time series analysis across various scenarios, achieved by fine-tuning a pre-trained language model, specifically GPT2, for the time series domain.

- LLMTime (Gruver et al. 2024) A zero-shot time series forecasting method that labels time series data as discrete numbers and converts the discrete labeled distributions into flexible density distributions over continuous values.

- Time-LLM (Jin et al. 2023) A reprogramming framework that repurposes large language models (LLMs) for general time series forecasting by aligning time series data with natural language modalities through text prototypes and Prompt-as-Prefix (PaP) techniques.

Evaluation Metrics For evaluation metrics in forecasting and imputation tasks, we utilize the mean square error (MSE) and mean absolute error (MAE) for long-term forecasting. In terms of the short-term forecasting on M4 benchmark, we adopt the symmetric mean absolute percentage error (SMAPE), mean absolute scaled error (MASE), and overall weighted average (OWA) as in N-BEATS. Note that OWA is a specific metric utilized in the M4 competition. The calculations of these metrics are as follows:

$$
\begin{aligned}
\mathrm{MSE} &= \frac{1}{H}\sum_{h=1}^{T}(\mathbf{Y}_h - \hat{\mathbf{Y}}_h)^2, \\
\mathrm{MAE} &= \frac{1}{H}\sum_{h=1}^{H}|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|, \\
\mathrm{SMAPE} &= \frac{200}{H}\sum_{h=1}^{H}\frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h| + |\hat{\mathbf{Y}}_h|}, \\
\mathrm{MAPE} &= \frac{100}{H}\sum_{h=1}^{H}\frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h|}, \\
\mathrm{MASE} &= \frac{1}{H}\sum_{h=1}^{H}\frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{\frac{1}{H-s}\sum_{j=s+1}^{H}|\mathbf{Y}_j - \mathbf{Y}_{j-s}|}, \\
\mathrm{OWA} &= \frac{1}{2}\left[\frac{\mathrm{SMAPE}}{\mathrm{SMAPE}_{\text{Naïve2}}} + \frac{\mathrm{MASE}}{\mathrm{MASE}_{\text{Naïve2}}}\right],
\end{aligned}
\tag{3}
$$

where s is the periodicity of the time series data. H denotes the number of data points (i.e., prediction horizon in our cases). $\mathbf{Y}_h$ and $\hat{\mathbf{Y}}_h$ are the h-th ground truth and prediction where $h \in \{1, \cdots, H\}$.

In addition, we used Precision (P), Recall (R), and F1-Score (F1) to simply quantify the performance of our FFTS and baselines on the anomaly detection task, these can be formulated as:

$$
\begin{aligned}
\mathrm{P} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \\
\mathrm{R} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \\
\mathrm{F1} &= 2 \times \frac{\mathrm{P} \times \mathrm{R}}{\mathrm{P}+\mathrm{R}},
\end{aligned}
\tag{4}
$$

where TP (True Positives), FP (False Positives), and FN (False Negatives) represent the number of samples correctly labeled as anomalous, the number of samples incorrectly labeled as anomalous, and the number of samples that were not labeled as anomalous by the model but were actually anomalous, respectively.

C. Theorem and Proofs

Theorem 1. *Consider a federated learning-based time series pretrianing system with m clients. Let $\mathcal{D}_1, \mathcal{D}_2, ..., \mathcal{D}_m$*

be the true data distribution and $\hat{\mathcal{D}}_1, \hat{\mathcal{D}}_2, ..., \hat{\mathcal{D}}_m$ be the empirical data distribution. Denote the head h as the hypothesis from $\mathcal{H}$ and d be the VC-dimension of $\mathcal{H}$. The total number of samples over all clients is N. Then with probability at least $1-\delta$:

$$\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left| \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right| \leq \sqrt{\frac{N}{2} \log \frac{(m+1)|\{\boldsymbol{P}\}|}{\delta}} + \sqrt{\frac{d}{N} \log \frac{eN}{d}}. \tag{5}$$

Proof. We start from the McDiarmid's inequality as

$$\mathbb{P}[g(X_1, ..., X_n) - \mathbb{E}[g(X_1, ..., X_n)] \geq \epsilon] \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2}\right) \tag{6}$$

when

$$\sup_{x_1,...,x_n} |g(x_1, x_2, ..., x_n) - g(x_1, x_2, ..., x_n)| \leq c_i \tag{7}$$

Eq. 6 equals to

$$\mathbb{P}[g(\cdot) - \mathbb{E}[g(\cdot)] \leq \epsilon] \geq 1 - \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2}\right) \tag{8}$$

which means that with probability at least $1 - \exp(-\frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2})$,

$$g(\cdot) - \mathbb{E}[g(\cdot)] \leq \epsilon \tag{9}$$

Let $\delta = \exp(-\frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2})$, the above can be rewritten as with parameters of adaptive trend-awareness module at least $1-\delta$,

$$g(\cdot) - \mathbb{E}[g(\cdot)] \leq \sqrt{\frac{\sum_{i=1}^{n} c_i^2}{2} \log \frac{1}{\delta}} \tag{10}$$

Now we substitute $g(\cdot)$ with our parameters of adaptive trend-awareness module as

$$\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left(\sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \tag{11}$$

we can obtain that with probability at least $1-\delta$, the following holds for specific parameters of adaptive trend-awareness module,

$$\begin{aligned} &\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left(\sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \\ &- \mathbb{E}\left[\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left(\sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \right] \\ &\leq \sqrt{\frac{N}{2} \log \frac{1}{\delta}} \end{aligned} \tag{12}$$

Considering there are $(m+1)|\{\boldsymbol{P}\}|$ adaptive trend-awareness module in total, by using Boole's inequality, with probability at least $1-\delta$, the following holds,

$$\begin{aligned} &\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left(\sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \\ &\leq \mathbb{E}\left[\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left(\sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \right] \\ &+ \sqrt{\frac{N}{2} \log \frac{(m+1)|\{\boldsymbol{P}\}|}{\delta}} \end{aligned} \tag{13}$$

where N is the total number of samples over all clients.

$$\begin{aligned} &\mathbb{E}\left[\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left(\sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \right] \\ &\leq \mathbb{E}\left[\sum_{i=1}^{m} \frac{|D_i|}{N} \max_{\{\boldsymbol{P}_i\}} \left(\mathcal{L}_{\mathcal{D}_i} - \mathcal{L}_{\hat{\mathcal{D}}_i} \right) \right] \\ &\leq^a \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{R}(\mathcal{H}) \\ &\leq \sum_{i=1}^{m} \frac{|D_i|}{N} \sqrt{\frac{d}{|D_i|} \log \frac{e|D_i|}{d}} \\ &\leq \sum_{i=1}^{m} \frac{D_i}{N} \sqrt{\frac{d}{|D_i|} \log \frac{eN}{d}} \\ &\leq^b \sqrt{\frac{d}{N} \log \frac{eN}{d}} \end{aligned} \tag{14}$$

where $\mathcal{H}$ is the hypothesis set of head h, d is the VC-dimension of $\mathcal{H}$. The a follow from the definition of Rademacher complexity

$$\mathcal{R}_n(\mathcal{F}) = \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i f(x_i) \right], \tag{15}$$

where $\sigma_1, \sigma_2, \ldots, \sigma_n$ are independent Rademacher random variables that take values in $\{-1, 1\}$ with equal probability, $\mathbb{E}_\sigma$ denotes the expectation over the Rademacher variables, $x_1, x_2, \ldots, x_n$ are the input data points, and the b follows from Jensen's inequality, so

$$\max_{(\{\boldsymbol{P}_1\},\{\boldsymbol{P}_2\},...,\{\boldsymbol{P}_m\})} \left| \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\mathcal{D}_i} - \sum_{i=1}^{m} \frac{|D_i|}{N} \mathcal{L}_{\hat{\mathcal{D}}_i} \right| \leq \sqrt{\frac{N}{2} \log \frac{(m+1)|\{\boldsymbol{P}\}|}{\delta}} + \sqrt{\frac{d}{N} \log \frac{eN}{d}} \tag{16}$$

□

D. Full Experiment Results

This section shows the results of the experiments that are missing from the main body. The full results are indexed as follows:

- Full results of long-term forecasting (normal), **Table 6** and **Table 7**.
- Full results of long-term forecasting (few-shot), **Table 8** and **Table 9**.
- Full results of long-term forecasting (zero-shot), **Table 10**.
- Full results of long-term forecasting (with additional baselines), **Table 11**.
- Full results of short-term forecasting, **Table 12** and **Table 13**.
- Full results of imputation, **Table 14**.
- Full results of anomaly detection, **Table 15**.

Dataset	Method / Metric	FFTS MSE	FFTS MAE	Time-LLM MSE	Time-LLM MAE	GPT4TS MSE	GPT4TS MAE	DLinear MSE	DLinear MAE	PatchTST MSE	PatchTST MAE	TimesNet MSE	TimesNet MAE
ETTh1	96	**0.342**	**0.378**	0.362	0.392	**0.376**	**0.397**	0.375	0.399	0.370	0.399	0.384	0.402
	192	**0.378**	**0.402**	0.398	**0.418**	0.416	**0.418**	**0.405**	0.416	0.413	0.421	0.436	0.429
	336	**0.412**	**0.422**	0.430	0.427	0.442	**0.433**	0.439	0.443	**0.422**	**0.436**	0.491	0.469
	720	**0.424**	**0.447**	0.442	**0.457**	0.477	0.456	0.472	0.490	**0.447**	0.466	0.521	0.500
	Avg.	**0.389**	**0.412**	0.408	0.423	0.465	0.455	**0.422**	**0.437**	**0.413**	0.430	0.458	0.450
ETTh2	96	**0.261**	**0.319**	0.268	0.328	**0.285**	**0.342**	0.289	0.353	0.274	0.336	0.340	0.374
	192	**0.322**	0.377	0.329	**0.375**	**0.354**	**0.389**	0.383	0.418	0.339	0.379	0.402	0.414
	336	0.354	0.402	**0.368**	**0.409**	0.373	0.407	0.448	0.465	**0.329**	**0.380**	0.452	0.452
	720	0.377	**0.412**	**0.372**	0.420	**0.406**	**0.441**	0.605	0.551	0.379	0.422	0.462	0.468
	Avg	**0.329**	**0.378**	0.334	0.383	0.381	0.412	0.431	0.446	**0.330**	**0.379**	0.414	0.427
ETTm1	96	**0.254**	**0.331**	0.272	0.334	**0.292**	0.346	0.299	**0.343**	0.290	0.342	0.338	0.375
	192	**0.301**	**0.343**	0.310	0.358	**0.332**	0.372	0.335	**0.365**	0.332	0.369	0.374	0.387
	336	**0.333**	**0.380**	0.352	0.384	**0.366**	**0.394**	0.369	0.386	0.366	0.392	0.410	0.411
	720	**0.370**	**0.400**	0.383	0.411	**0.417**	**0.421**	0.425	0.421	0.416	0.420	0.478	0.450
	Avg	**0.315**	**0.364**	0.329	0.372	0.388	0.403	**0.357**	**0.378**	0.351	0.380	0.400	0.406
ETTm2	96	**0.150**	**0.241**	0.161	0.253	**0.173**	**0.262**	0.167	0.269	0.165	0.255	0.187	0.267
	192	0.227	0.294	**0.219**	**0.293**	0.229	0.301	0.224	0.303	**0.220**	**0.292**	0.249	0.309
	336	**0.254**	**0.318**	0.271	0.329	**0.286**	**0.341**	0.281	0.342	0.274	0.329	0.321	0.351
	720	**0.346**	**0.376**	0.352	0.379	**0.378**	**0.401**	0.397	0.421	0.362	0.385	0.408	0.403
	Avg	**0.244**	**0.307**	0.251	0.313	0.284	0.339	**0.267**	**0.333**	0.255	0.315	0.291	0.333
Weather	96	**0.141**	0.203	0.147	**0.201**	**0.162**	**0.212**	0.176	0.237	0.149	0.198	0.172	0.220
	192	**0.178**	**0.230**	0.189	0.234	**0.204**	**0.248**	0.220	0.282	0.194	0.241	0.219	0.261
	336	**0.241**	**0.281**	**0.262**	0.279	0.254	**0.286**	0.265	0.319	0.245	0.282	0.280	0.306
	720	**0.300**	**0.312**	0.304	0.316	**0.326**	**0.337**	0.333	0.362	0.314	0.334	0.365	0.359
	Avg	**0.215**	**0.256**	0.225	0.257	**0.237**	**0.270**	0.248	0.300	0.225	0.264	0.259	0.287
ILI	24	1.231	0.771	**1.285**	**0.727**	**2.063**	**0.881**	2.215	1.081	1.319	0.754	2.317	0.934
	36	**1.321**	0.798	1.404	**0.814**	**1.868**	**0.892**	1.963	0.963	1.430	0.834	1.972	0.920
	48	**1.412**	**0.787**	1.523	0.807	**1.790**	**0.884**	2.130	1.024	1.553	0.815	2.238	0.940
	60	**1.428**	0.808	1.531	0.854	**1.979**	0.957	2.368	1.096	1.470	**0.788**	2.027	**0.928**
	Avg	**1.348**	**0.791**	1.435	0.801	**1.925**	**0.903**	2.169	1.041	1.443	0.797	2.139	0.931

Dataset	Method / Metric	FEDformer MSE	FEDformer MAE	Autoformer MSE	Autoformer MAE	Stationary MSE	Stationary MAE	ETSformer MSE	ETSformer MAE	LightTS MSE	LightTS MAE	informer MSE	informer MAE	Reformer MSE	Reformer MAE
ETTh1	96	0.376	0.419	0.449	0.459	0.513	0.491	0.494	0.479	0.424	0.432	0.865	0.713	0.837	0.728
	192	0.420	0.448	0.500	0.482	0.534	0.504	0.538	0.504	0.475	0.462	1.008	0.792	0.923	0.766
	336	0.459	0.465	0.521	0.496	0.588	0.535	0.574	0.521	0.518	0.488	1.107	0.809	1.097	0.835
	720	0.506	0.507	0.514	0.512	0.643	0.616	0.562	0.535	0.547	0.533	1.181	0.865	1.257	0.889
	Avg.	0.440	0.460	0.496	0.487	0.570	0.537	0.542	0.510	0.491	0.479	1.040	0.795	1.029	0.805
ETTh2	96	0.358	0.397	0.346	0.388	0.476	0.458	0.340	0.391	0.397	0.437	3.755	1.525	2.626	1.317
	192	0.429	0.439	0.456	0.452	0.512	0.493	0.430	0.439	0.520	0.504	5.602	1.931	11.12	2.979
	336	0.496	0.487	0.482	0.486	0.552	0.551	0.485	0.479	0.626	0.559	4.721	1.835	9.323	2.769
	720	0.463	0.474	0.515	0.511	0.562	0.560	0.500	0.497	0.863	0.672	3.647	1.625	3.874	1.697
	Avg	0.437	0.449	0.450	0.459	0.526	0.516	0.439	0.452	0.602	0.543	4.431	1.729	6.736	2.191
ETTm1	96	0.379	0.419	0.505	0.475	0.386	0.398	0.375	0.398	0.374	0.400	0.672	0.571	0.538	0.528
	192	0.426	0.441	0.553	0.496	0.459	0.444	0.408	0.410	0.400	0.407	0.795	0.669	0.658	0.592
	336	0.445	0.459	0.621	0.537	0.495	0.464	0.435	0.428	0.438	0.438	1.212	0.871	0.898	0.721
	720	0.543	0.490	0.671	0.561	0.585	0.516	0.499	0.462	0.527	0.502	1.166	0.823	1.102	0.841
	Avg	0.448	0.452	0.588	0.517	0.481	0.456	0.429	0.425	0.435	0.437	0.961	0.734	0.799	0.671
ETTm2	96	0.203	0.287	0.255	0.339	0.192	0.274	0.189	0.280	0.209	0.308	0.365	0.453	0.658	0.619
	192	0.269	0.328	0.281	0.340	0.280	0.339	0.253	0.319	0.311	0.382	0.533	0.563	1.078	0.827
	336	0.325	0.366	0.339	0.372	0.334	0.361	0.314	0.357	0.442	0.466	1.363	0.887	1.549	0.972
	720	0.421	0.415	0.433	0.432	0.417	0.413	0.414	0.413	0.675	0.587	3.379	1.338	2.631	1.242
	Avg	0.305	0.349	0.327	0.371	0.306	0.347	0.293	0.342	0.409	0.436	1.410	0.810	1.479	0.915
Weather	96	0.217	0.296	0.266	0.336	0.173	0.223	0.197	0.281	0.182	0.242	0.300	0.384	0.689	0.596
	192	0.276	0.336	0.307	0.367	0.245	0.285	0.237	0.312	0.227	0.287	0.598	0.544	0.752	0.638
	336	0.339	0.380	0.359	0.395	0.321	0.338	0.298	0.353	0.282	0.334	0.578	0.523	0.639	0.596
	720	0.403	0.428	0.419	0.428	0.414	0.410	0.352	0.288	0.352	0.386	1.059	0.741	1.130	0.792
	Avg	0.309	0.360	0.338	0.382	0.288	0.314	0.271	0.334	0.261	0.312	0.634	0.548	0.803	0.656
ILI	24	3.228	1.260	3.483	1.287	2.294	0.945	2.527	1.020	8.313	2.144	5.764	1.677	4.400	1.382
	36	2.679	1.080	3.103	1.148	1.825	0.848	2.615	1.007	6.631	1.902	4.755	1.467	4.783	1.448
	48	2.622	1.078	2.669	1.085	2.010	0.900	2.359	0.972	7.299	1.982	4.763	1.469	4.832	1.465
	60	2.857	1.157	2.770	1.125	2.178	0.963	2.487	1.016	7.283	1.985	5.264	1.564	4.882	1.483
	Avg	2.847	1.144	3.006	1.161	2.077	0.914	2.497	1.004	7.382	2.003	5.137	1.544	4.724	1.445

Table 6: Long-term forecasting results. **Bold**: the best, Underline: the second best, **Blue**: the best in deep time series model baseline.

Setting		FFTS				FL Baseline					
Dataset	Method / Metric	FFTS (FT) MSE	FFTS (FT) MAE	FFTS-Cen (FT) MSE	FFTS-Cen (FT) MAE	FedAvg MSE	FedAvg MAE	FedProx MSE	FedProx MAE	pFedMe MSE	pFedMe MAE
ETTh1	96	**0.342**	0.378	0.345	**0.374**	0.354	0.381	0.388	0.409	0.387	0.405
	192	**0.378**	0.402	**0.379**	**0.397**	0.390	0.411	0.410	0.433	0.412	0.429
	336	**0.412**	0.422	0.416	**0.417**	0.429	0.430	0.449	0.451	0.448	0.454
	720	**0.424**	**0.447**	0.428	0.450	0.431	0.451	0.456	0.460	0.452	0.467
	Avg.	**0.389**	0.412	0.392	**0.410**	0.400	0.422	0.425	0.440	0.425	0.439
ETTh2	96	**0.261**	**0.319**	0.265	0.324	0.273	0.320	0.290	0.332	0.287	0.329
	192	**0.322**	0.377	0.326	**0.375**	0.331	0.380	0.349	0.404	0.350	0.408
	336	**0.354**	**0.402**	0.356	0.404	0.363	0.411	0.377	0.423	0.374	0.415
	720	**0.377**	0.412	0.378	**0.409**	0.384	0.427	0.400	0.430	0.399	0.427
	Avg.	**0.329**	**0.378**	0.331	0.378	0.342	0.385	0.354	0.394	0.353	0.389
ETTm1	96	**0.254**	0.331	0.256	0.333	0.269	0.333	0.295	0.342	0.299	0.354
	192	**0.301**	0.343	0.303	0.347	0.320	0.350	0.350	0.372	0.354	0.369
	336	**0.333**	0.380	0.327	0.378	0.357	0.388	0.394	0.400	0.392	0.401
	720	0.370	0.400	**0.368**	0.401	0.394	0.408	0.441	0.437	0.442	0.441
	Avg.	0.315	**0.364**	**0.314**	0.365	0.335	0.370	0.370	0.390	0.372	0.391
ETTm2	96	**0.150**	**0.241**	0.153	0.245	0.162	0.253	0.188	0.278	0.186	0.272
	192	**0.227**	**0.294**	0.229	0.300	0.229	0.300	0.258	0.317	0.256	0.320
	336	**0.254**	**0.318**	0.256	0.320	0.265	0.322	0.337	0.370	0.332	0.364
	720	0.346	**0.376**	**0.343**	0.379	0.350	0.377	0.435	0.430	0.433	0.427
	Avg.	**0.244**	**0.307**	0.245	0.311	0.251	0.313	0.309	0.348	0.302	0.346
Weather	96	**0.141**	**0.203**	0.143	0.206	0.145	0.204	0.178	0.233	0.180	0.230
	192	0.178	**0.230**	**0.174**	0.232	0.180	0.232	0.230	0.271	0.227	0.276
	336	**0.241**	**0.281**	0.243	0.284	0.243	0.285	0.288	0.310	0.286	0.303
	720	**0.300**	**0.312**	0.304	0.312	0.304	0.314	0.354	0.344	0.347	0.335
	Avg.	**0.215**	**0.256**	0.217	0.259	0.218	0.259	0.263	0.290	0.260	0.286
ILI	24	**1.231**	**0.771**	1.240	0.775	1.245	0.780	1.414	0.841	1.435	0.829
	36	**1.321**	**0.798**	1.335	0.800	1.345	0.801	1.587	0.883	1.561	0.890
	48	1.412	**0.787**	**1.410**	0.792	1.487	0.799	1.599	0.901	1.629	0.892
	60	1.428	**0.808**	**1.427**	0.812	1.480	0.834	1.600	0.942	1.587	0.923
	Avg.	**1.348**	**0.791**	1.353	0.795	1.390	0.804	1.550	0.892	1.553	0.884

Table 7: Long-term forecasting based on FL. **Bold**: the best, Underline: the second best.

Setting		FFTS				Deep Time Series Model Baseline																											
Dataset	Method	FFTS (FT)		FFTS-Cen (FT)		Time-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		informer		Reformer	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.465**	**0.452**	0.467	0.455	**0.483**	**0.464**	0.543	0.506	0.547	0.503	0.557	0.519	0.892	0.625	0.593	0.529	0.681	0.570	0.952	0.650	1.169	0.832	1.483	0.91	1.225	0.812	1.198	0.795
	192	**0.610**	0.541	0.612	**0.543**	**0.629**	**0.540**	0.748	0.580	0.720	0.604	0.711	0.570	0.940	0.665	0.652	0.563	0.725	0.602	0.943	0.645	1.221	0.853	1.525	0.93	1.249	0.828	1.273	0.853
	336	**0.744**	0.576	0.749	**0.572**	**0.768**	0.626	0.754	0.595	0.984	0.727	0.816	0.619	0.945	0.653	0.731	**0.594**	0.761	0.624	0.935	0.644	1.179	0.832	1.347	0.87	1.202	0.811	1.254	0.857
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	**0.606**	**0.523**	0.609	**0.523**	**0.627**	**0.543**	0.681	0.560	0.750	0.611	0.694	0.569	0.925	0.647	0.658	0.562	0.722	0.598	0.943	0.646	1.189	0.839	1.451	0.903	1.225	0.817	1.241	0.835
ETTh2	96	0.321	0.386	**0.317**	**0.385**	**0.336**	**0.397**	0.376	0.421	0.442	0.456	0.401	0.421	0.409	0.420	0.390	0.424	0.428	0.468	0.408	0.423	0.678	0.619	2.022	1.006	3.837	1.508	3.753	1.518
	192	**0.382**	**0.412**	0.386	0.415	**0.406**	**0.425**	0.418	0.441	0.617	0.542	0.452	0.455	0.483	0.464	0.457	0.465	0.496	0.504	0.497	0.468	0.845	0.697	3.534	1.348	3.975	1.933	3.516	1.473
	336	**0.387**	**0.426**	0.389	0.427	**0.405**	**0.432**	0.408	0.439	1.424	0.849	0.464	0.469	0.499	0.479	0.477	0.483	0.486	0.496	0.507	0.481	0.905	0.727	4.063	1.451	3.956	1.520	3.312	1.427
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	**0.363**	**0.408**	0.365	0.409	**0.382**	**0.418**	0.400	0.433	0.694	0.577	0.827	0.615	0.439	0.448	0.463	0.454	0.441	0.457	0.470	0.489	0.809	0.681	3.206	1.268	3.922	1.653	3.527	1.472
ETTm1	96	0.337	0.374	**0.335**	**0.372**	**0.316**	**0.377**	0.386	0.405	0.332	0.374	0.399	0.414	0.606	0.518	0.628	0.544	0.726	0.578	0.823	0.587	1.031	0.747	1.048	0.733	1.130	0.775	1.234	0.798
	192	0.412	0.408	**0.423**	**0.413**	0.450	0.464	0.440	0.438	**0.358**	**0.390**	0.441	0.436	0.681	0.539	0.666	0.566	0.750	0.591	0.844	0.591	1.087	0.766	1.097	0.756	1.150	0.788	1.287	0.839
	336	**0.436**	0.411	**0.443**	**0.417**	0.450	**0.424**	0.485	0.459	0.402	0.416	0.499	0.467	0.786	0.597	0.807	0.628	0.851	0.659	0.870	0.603	1.138	0.787	1.147	0.775	1.198	0.809	1.288	0.842
	720	**0.457**	**0.460**	0.464	0.461	**0.483**	**0.471**	0.577	0.499	0.511	0.489	0.767	0.587	0.796	0.593	0.822	0.633	0.857	0.655	0.893	0.611	1.245	0.831	1.200	0.799	1.175	0.794	1.247	0.828
	Avg	**0.411**	0.413	0.417	**0.416**	**0.425**	**0.434**	0.472	0.450	0.400	0.417	0.526	0.476	0.717	0.561	0.730	0.592	0.796	0.620	0.857	0.598	1.125	0.782	1.123	0.765	1.163	0.791	1.264	0.826
ETTm2	96	**0.152**	0.253	0.153	**0.250**	**0.174**	**0.261**	0.199	0.280	0.236	0.326	0.206	0.288	0.220	0.299	0.229	0.320	0.232	0.322	0.238	0.316	0.404	0.485	1.108	0.772	3.599	1.478	3.883	1.545
	192	**0.202**	**0.266**	0.205	0.268	**0.215**	**0.287**	0.256	0.316	0.306	0.373	0.264	0.324	0.311	0.361	0.394	0.361	0.291	0.357	0.298	0.349	0.479	0.521	1.317	0.850	3.578	1.475	3.553	1.484
	336	**0.259**	0.312	0.263	**0.309**	**0.273**	**0.330**	0.318	0.353	0.380	0.423	0.334	0.367	0.338	0.366	0.378	0.427	0.478	0.517	0.353	0.380	0.552	0.555	1.415	0.879	3.561	1.473	3.446	1.460
	720	**0.421**	0.400	0.427	**0.398**	**0.433**	**0.412**	0.460	0.436	0.674	0.583	0.454	0.432	0.509	0.465	0.523	0.510	0.553	0.538	0.475	0.445	0.701	0.627	1.822	0.984	3.896	1.533	3.445	1.460
	Avg	**0.259**	0.308	0.263	**0.306**	**0.274**	**0.323**	0.308	0.346	0.399	0.426	0.314	0.352	0.344	0.372	0.381	0.404	0.388	0.433	0.341	0.372	0.534	0.547	1.415	0.871	3.658	1.489	3.581	1.487
Weather	96	**0.161**	**0.265**	0.164	0.263	0.172	0.263	0.175	**0.230**	0.184	0.242	**0.171**	0.224	0.207	0.253	0.229	0.309	0.227	0.299	0.215	0.252	0.218	0.295	0.230	0.285	0.497	0.497	0.406	0.435
	192	**0.201**	0.256	0.203	**0.252**	**0.224**	**0.271**	0.227	0.276	0.228	0.283	0.230	0.277	0.272	0.307	0.265	0.317	0.278	0.333	0.290	0.307	0.294	0.331	0.274	0.323	0.620	0.545	0.446	0.450
	336	0.268	**0.300**	**0.267**	0.301	**0.282**	**0.321**	0.286	0.322	0.279	0.322	0.294	0.326	0.313	0.328	0.353	0.392	0.351	0.393	0.353	0.348	0.359	0.398	0.318	0.355	0.649	0.547	0.465	0.459
	720	**0.348**	0.386	0.349	**0.383**	**0.366**	**0.381**	0.366	0.379	0.364	0.388	0.384	0.387	0.400	0.385	0.391	0.394	0.387	0.389	0.452	0.407	0.461	0.461	0.401	0.418	0.570	0.522	0.471	0.468
	Avg	**0.245**	0.302	0.247	**0.300**	**0.260**	**0.309**	0.263	0.301	0.263	0.308	0.269	0.303	0.298	0.318	0.309	0.353	0.310	0.353	0.327	0.328	0.333	0.371	0.305	0.345	0.584	0.527	0.447	0.453

Table 8: Long-term forecasting results on 5% few-shot learning. **Bold**: the best, Underline: the second best, **Blue**: the best in deep time series model baseline.

Setting		FFTS				Deep Time Series Model Baseline																											
Dataset	Method	FFTS (FT)		FFTS-Cen (FT)		Time-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		informer		Reformer	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.432**	**0.448**	0.440	0.446	**0.448**	0.460	0.458	0.456	0.492	0.495	0.516	0.485	0.861	0.628	0.512	0.499	0.613	0.552	0.918	0.639	1.112	0.806	1.298	0.838	1.179	0.792	1.184	0.790
	192	0.465	**0.467**	**0.462**	0.470	**0.484**	0.483	0.570	0.516	0.565	0.538	0.598	0.524	0.797	0.593	0.624	0.555	0.722	0.598	0.915	0.629	1.155	0.823	1.322	0.854	1.199	0.806	1.295	0.850
	336	0.592	0.537	0.594	**0.535**	**0.589**	**0.540**	0.608	**0.535**	0.721	0.622	0.657	0.550	0.941	0.648	0.691	0.574	0.750	0.619	0.939	0.644	1.179	0.832	1.347	0.870	1.202	0.811	1.294	0.854
	720	**0.695**	0.600	0.701	**0.610**	**0.700**	0.604	0.725	**0.591**	0.986	0.743	0.762	0.610	0.877	0.641	0.728	0.614	0.721	0.616	0.887	0.645	1.273	0.874	1.534	0.947	1.217	0.825	1.223	0.838
	Avg	**0.546**	**0.513**	0.549	0.515	**0.556**	0.522	0.590	0.525	0.691	0.600	0.633	0.542	0.869	0.628	0.639	0.561	0.702	0.596	0.915	0.639	1.180	0.834	1.375	0.877	1.199	0.809	1.249	0.833
ETTh2	96	**0.262**	0.334	0.264	**0.332**	**0.275**	**0.326**	0.331	0.374	0.357	0.411	0.353	0.389	0.378	0.409	0.382	0.416	0.413	0.451	0.389	0.411	0.678	0.619	2.022	1.006	3.837	1.508	3.788	1.533
	192	0.365	0.370	**0.362**	**0.362**	**0.374**	**0.373**	0.402	0.411	0.569	0.519	0.403	0.414	0.490	0.467	0.478	0.474	0.474	0.477	0.473	0.455	0.785	0.666	2.329	1.104	3.856	1.513	3.552	1.483
	336	**0.402**	0.433	0.404	**0.430**	**0.406**	**0.429**	0.406	0.433	0.671	0.572	0.426	0.441	0.537	0.494	0.504	0.501	0.547	0.543	0.507	0.480	0.839	0.694	2.453	1.122	3.952	1.526	3.395	1.526
	720	**0.420**	**0.437**	0.421	0.440	**0.427**	**0.449**	0.449	0.464	0.824	0.648	0.477	0.480	0.510	0.491	0.499	0.509	0.516	0.523	0.477	0.472	1.273	0.874	3.816	1.407	3.842	1.503	3.205	1.401
	Avg	**0.362**	0.394	0.363	**0.391**	**0.370**	**0.394**	0.397	0.421	0.605	0.538	0.415	0.431	0.479	0.465	0.466	0.475	0.488	0.499	0.462	0.455	0.894	0.713	2.655	1.160	3.872	1.513	3.485	1.486
ETTm1	96	**0.331**	**0.376**	0.336	0.377	**0.346**	**0.388**	0.390	0.404	0.352	0.392	0.410	0.419	0.583	0.501	0.578	0.518	0.774	0.614	0.761	0.568	0.911	0.688	0.921	0.682	1.162	0.785	1.442	0.847
	192	0.361	**0.404**	**0.358**	**0.404**	**0.373**	**0.416**	0.429	0.423	0.382	0.412	0.437	0.434	0.630	0.528	0.617	0.546	0.754	0.592	0.781	0.574	0.955	0.703	0.957	0.701	1.172	0.793	1.444	0.862
	336	0.401	0.412	**0.398**	**0.408**	**0.413**	**0.426**	0.469	0.439	0.419	0.434	0.476	0.454	0.725	0.568	0.998	0.775	0.869	0.677	0.803	0.587	0.991	0.719	0.998	0.716	1.227	0.908	1.450	0.866
	720	**0.452**	**0.459**	0.454	0.460	**0.485**	**0.476**	0.569	0.498	0.490	0.477	0.681	0.556	0.769	0.549	0.693	0.579	0.810	0.630	0.844	0.581	1.062	0.747	1.007	0.719	1.207	0.797	1.366	0.850
	Avg	**0.386**	0.413	0.387	**0.412**	**0.404**	**0.427**	0.464	0.441	0.411	0.429	0.501	0.466	0.677	0.537	0.722	0.605	0.802	0.628	0.797	0.578	0.980	0.714	0.971	0.705	1.192	0.821	1.426	0.856
ETTm2	96	0.174	0.255	**0.173**	**0.254**	**0.177**	**0.261**	0.188	0.269	0.213	0.303	0.191	0.274	0.212	0.285	0.291	0.399	0.352	0.454	0.229	0.308	0.331	0.430	0.813	0.688	3.203	1.407	4.195	1.628
	192	**0.229**	**0.302**	0.232	0.304	**0.241**	0.314	0.251	**0.309**	0.278	0.345	0.252	0.317	0.270	0.323	0.307	0.379	0.694	0.691	0.291	0.343	0.400	0.464	1.008	0.768	3.112	1.387	4.042	1.601
	336	**0.256**	**0.311**	0.265	0.314	**0.274**	**0.327**	0.307	0.346	0.338	0.385	0.306	0.353	0.323	0.353	0.543	0.559	2.408	1.407	0.348	0.376	0.469	0.498	1.031	0.775	3.255	1.421	3.963	1.585
	720	**0.404**	0.392	0.407	**0.384**	**0.417**	**0.390**	0.426	0.417	0.436	0.440	0.433	0.427	0.474	0.449	0.712	0.614	1.913	1.166	0.461	0.438	0.589	0.557	1.096	0.791	3.909	1.543	3.711	1.532
	Avg	**0.266**	0.315	0.270	**0.314**	**0.277**	**0.323**	0.293	0.335	0.316	0.368	0.296	0.343	0.320	0.353	0.463	0.488	1.342	0.930	0.332	0.366	0.447	0.487	0.987	0.756	3.370	1.440	3.978	1.587
Weather	96	**0.150**	**0.211**	0.154	0.212	**0.161**	**0.210**	0.163	0.215	0.171	0.224	0.165	0.215	0.184	0.230	0.188	0.253	0.221	0.297	0.192	0.234	0.199	0.272	0.217	0.269	0.374	0.401	0.335	0.380
	192	**0.195**	**0.223**	0.196	0.222	**0.204**	0.248	0.210	0.254	0.215	0.263	0.210	0.257	0.245	0.283	0.250	0.304	0.270	0.322	0.269	0.295	0.279	0.332	0.259	**0.304**	0.552	0.478	0.522	0.462
	336	**0.232**	**0.291**	0.234	0.290	0.261	**0.302**	0.256	**0.292**	0.258	0.299	0.259	0.297	0.305	0.321	0.312	0.346	0.320	0.351	0.370	0.357	0.356	0.386	**0.303**	0.334	724	0.541	0.715	0.535
	720	**0.288**	**0.316**	0.291	0.320	**0.309**	**0.332**	0.321	0.339	0.320	0.346	0.332	0.346	0.381	0.371	0.387	0.393	0.390	0.396	0.441	0.405	0.437	0.448	0.377	0.382	0.739	0.558	0.611	0.500
	Avg	**0.216**	**0.260**	0.219	0.261	**0.234**	**0.273**	0.238	0.275	0.241	0.283	0.242	0.279	0.279	0.301	0.284	0.324	0.300	0.342	0.318	0.323	0.318	0.360	0.289	0.322	0.597	0.495	0.546	0.469

Table 9: Long-term forecasting results on 10% few-shot learning. **Bold**: the best, Underline: the second best, **Blue**: the best in deep time series model baseline.

Setting		FFTS				Deep Time Series Model Baseline															
Methods		FFTS (FT)		FFTS-Cen (ft)		Time-LLM		LLMTime		GPT4TS		DLinear		PatchTST		TimesNet		Autoformer	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1 - ETTh2	96	**0.272**	0.328	0.276	**0.322**	**0.279**	**0.337**	0.510	0.576	0.335	0.374	0.347	0.400	0.304	0.350	0.358	0.387	0.469	0.486
	192	**0.333**	**0.362**	0.337	0.368	**0.351**	**0.374**	0.523	0.586	0.412	0.417	0.447	0.460	0.386	0.400	0.427	0.429	0.634	0.567
	336	**0.371**	0.404	0.377	**0.400**	**0.388**	**0.415**	0.640	0.637	0.441	0.444	0.515	0.505	0.414	0.428	0.449	0.451	0.655	0.588
	720	**0.379**	**0.410**	0.383	0.412	**0.391**	**0.420**	2.296	1.034	0.438	0.452	0.665	0.589	0.419	0.443	0.448	0.458	0.570	0.549
	Avg	**0.339**	**0.376**	0.343	**0.376**	**0.353**	**0.387**	0.992	0.708	0.406	0.422	0.493	0.488	0.380	0.405	0.421	0.431	0.582	0.548
ETTh1 - ETTm2	96	0.187	0.290	**0.184**	**0.288**	**0.189**	**0.293**	0.646	0.563	0.236	0.315	0.255	0.357	0.215	0.304	0.239	0.313	0.352	0.432
	192	0.230	**0.303**	**0.231**	0.307	**0.237**	**0.312**	0.934	0.654	0.287	0.342	0.338	0.413	0.275	0.339	0.291	0.342	0.413	0.460
	336	**0.288**	**0.354**	0.295	0.358	**0.291**	**0.365**	1.157	0.728	0.341	0.374	0.425	0.465	0.334	0.373	0.342	0.371	0.465	0.489
	720	**0.360**	**0.380**	**0.364**	0.384	**0.372**	**0.390**	4.730	1.531	0.435	0.422	0.640	0.573	0.431	0.424	0.434	0.419	0.599	0.551
	Avg	**0.266**	**0.332**	**0.269**	0.334	**0.273**	**0.340**	1.867	0.869	0.325	0.363	0.415	0.452	0.314	0.360	0.327	0.361	0.457	0.483
ETTh2 - ETTh1	96	**0.430**	**0.441**	0.436	0.448	**0.450**	**0.452**	1.130	0.777	0.732	0.577	0.689	0.555	0.485	0.465	0.848	0.601	0.693	0.569
	192	**0.450**	**0.455**	0.454	0.450	**0.465**	**0.461**	1.242	0.820	0.758	0.559	0.707	0.568	0.565	0.509	0.860	0.610	0.760	0.601
	336	**0.478**	**0.468**	0.480	0.470	**0.501**	**0.482**	1.328	0.864	0.759	0.578	0.710	0.577	0.581	0.515	0.867	0.626	0.781	0.619
	720	0.483	**0.470**	**0.482**	0.471	**0.501**	**0.502**	4.145	1.461	0.781	0.597	0.704	0.596	0.628	0.561	0.887	0.648	0.796	0.644
	Avg	**0.460**	**0.459**	0.463	0.460	**0.479**	**0.474**	1.961	0.981	0.757	0.578	0.703	0.574	0.565	0.513	0.865	0.621	0.757	0.608
ETTh2 - ETTm2	96	0.168	0.263	**0.166**	**0.262**	**0.174**	**0.276**	0.646	0.563	0.253	0.329	0.240	0.336	0.226	0.309	0.248	0.324	0.263	0.352
	192	**0.231**	**0.300**	0.234	0.301	**0.233**	**0.315**	0.934	0.654	0.293	0.346	0.295	0.369	0.289	0.345	0.296	0.352	0.326	0.389
	336	0.276	**0.318**	**0.273**	**0.318**	**0.291**	**0.337**	1.157	0.728	0.347	0.376	0.345	0.397	0.348	0.379	0.353	0.383	0.387	0.426
	720	**0.378**	**0.405**	0.380	0.408	**0.392**	**0.417**	4.730	1.531	0.446	0.429	0.432	0.442	0.439	0.427	0.471	0.446	0.487	0.478
	Avg	**0.263**	**0.321**	**0.263**	0.322	**0.272**	**0.341**	1.867	0.869	0.335	0.370	0.328	0.386	0.325	0.365	0.342	0.376	0.366	0.411
ETTm1- ETTh2	96	0.303	0.354	**0.300**	**0.350**	**0.321**	**0.369**	0.510	0.576	0.353	0.392	0.365	0.415	0.354	0.385	0.377	0.407	0.435	0.470
	192	0.372	0.409	**0.369**	**0.404**	**0.389**	**0.410**	0.523	0.586	0.443	0.437	0.454	0.462	0.447	0.434	0.471	0.453	0.495	0.489
	336	0.400	0.427	**0.389**	**0.420**	**0.408**	**0.433**	0.640	0.637	0.469	0.461	0.496	0.494	0.481	0.463	0.472	0.484	0.470	0.472
	720	**0.402**	**0.430**	0.410	0.433	**0.406**	**0.436**	2.296	1.034	0.466	0.468	0.541	0.529	0.474	0.471	0.495	0.482	0.480	0.485
	Avg	0.369	0.405	**0.367**	**0.402**	**0.381**	**0.412**	0.992	0.708	0.433	0.439	0.464	0.475	0.439	0.438	0.457	0.454	0.470	0.479
ETTm1- ETTm2	96	**0.151**	0.240	0.154	0.242	**0.169**	0.257	0.646	0.563	0.217	0.294	0.221	0.314	0.195	**0.271**	0.222	0.295	0.385	0.457
	192	**0.209**	**0.311**	0.210	0.318	0.227	0.318	0.934	0.654	0.277	0.327	0.286	0.359	**0.258**	0.311	0.288	0.337	0.433	0.469
	336	**0.271**	**0.321**	0.276	0.329	**0.290**	0.338	1.157	0.728	0.331	0.360	0.357	0.406	0.317	0.348	0.341	0.367	0.476	0.477
	720	**0.364**	**0.359**	0.368	0.359	**0.375**	0.367	4.730	1.531	0.429	0.413	0.476	0.476	0.416	0.404	0.436	0.418	0.582	0.535
	Avg	**0.249**	**0.308**	0.254	0.313	**0.268**	0.320	1.867	0.869	0.313	0.348	0.335	0.389	0.296	0.334	0.322	0.354	0.469	0.484
ETTm2- ETTh2	96	0.278	**0.354**	**0.276**	**0.354**	**0.298**	0.356	0.510	0.576	0.360	0.401	0.333	0.391	0.327	0.367	0.360	0.401	0.353	0.393
	192	**0.344**	0.392	0.358	**0.390**	**0.359**	**0.397**	0.523	0.586	0.434	0.437	0.441	0.456	0.411	0.418	0.434	0.437	0.432	0.437
	336	**0.351**	**0.399**	0.364	0.400	**0.367**	**0.412**	0.640	0.637	0.460	0.459	0.505	0.503	0.439	0.447	0.460	0.459	0.452	0.459
	720	0.383	0.423	**0.382**	**0.420**	**0.393**	**0.434**	2.296	1.034	0.485	0.477	0.543	0.534	0.459	0.470	0.485	0.477	0.453	0.467
	Avg	**0.339**	0.392	0.345	**0.391**	**0.354**	**0.400**	0.992	0.708	0.435	0.443	0.455	0.471	0.409	0.425	0.435	0.443	0.423	0.439
ETTm2- ETTh1	96	0.339	0.398	**0.336**	**0.395**	**0.359**	**0.397**	1.179	0.781	0.747	0.558	0.570	0.490	0.491	0.437	0.747	0.558	0.735	0.576
	192	0.388	0.421	**0.386**	**0.429**	**0.390**	**0.420**	1.327	0.846	0.781	0.560	0.590	0.506	0.530	0.470	0.781	0.560	0.753	0.586
	336	**0.416**	**0.441**	0.419	0.444	**0.421**	**0.445**	1.478	0.902	0.778	0.578	0.706	0.567	0.565	0.497	0.778	0.578	0.750	0.593
	720	**0.468**	**0.482**	0.475	**0.482**	**0.487**	0.488	3.749	1.408	0.769	0.573	0.731	0.584	0.686	0.565	0.769	0.573	0.782	0.609
	Avg	**0.403**	**0.436**	0.405	0.437	**0.414**	**0.438**	1.933	0.984	0.769	0.567	0.649	0.537	0.568	0.492	0.769	0.567	0.755	0.591

Table 10: Zero-shot results on long-term forecasting task. **Bold**: the best, Underline: the second best, **Blue**: the best in deep time series model baseline.

Dataset	Method	FFTS (FT)		N-BEATS		N-HiTS		AutoARIMA		AutoTheta		AutoETS	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.342**	**0.378**	**0.496**	**0.475**	0.392	0.407	0.933	0.635	1.266	0.758	1.264	0.756
	192	**0.378**	**0.402**	**0.544**	**0.504**	0.442	0.438	0.868	0.621	1.188	0.749	1.181	0.745
	336	**0.412**	**0.422**	**0.592**	**0.533**	0.497	0.471	0.964	0.663	1.310	0.799	1.292	0.792
	720	**0.424**	**0.447**	**0.639**	**0.588**	0.559	0.533	1.043	0.705	1.510	0.882	1.405	0.842
	Avg	**0.389**	**0.412**	**0.568**	**0.525**	0.473	0.462	0.952	0.656	1.319	0.797	1.286	0.784
ETTh2	96	**0.261**	**0.319**	**0.384**	**0.431**	0.321	0.368	0.390	0.417	0.461	0.430	0.444	0.403
	192	**0.322**	**0.377**	**0.496**	**0.493**	0.398	0.421	0.545	0.492	0.754	0.537	0.771	0.461
	336	**0.354**	**0.402**	**0.585**	**0.542**	0.453	0.459	0.697	0.562	1.355	0.683	1.526	0.522
	720	**0.377**	**0.412**	**0.792**	**0.651**	0.775	0.609	0.907	0.658	3.971	1.061	5.183	0.633
	Avg	**0.329**	**0.378**	**0.564**	**0.529**	0.487	0.464	0.635	0.532	1.635	0.678	1.981	0.505
ETTm1	96	**0.254**	**0.331**	**0.393**	**0.412**	0.327	0.368	1.091	0.661	1.211	0.704	1.519	0.768
	192	**0.301**	**0.343**	**0.425**	**0.427**	0.376	0.400	1.119	0.682	1.237	0.724	1.535	0.784
	336	**0.333**	**0.380**	**0.464**	**0.454**	0.407	0.423	1.125	0.698	1.231	0.735	1.472	0.782
	720	**0.370**	**0.400**	**0.521**	**0.488**	0.471	0.456	1.243	0.745	1.394	0.801	1.591	0.825
	Avg	**0.315**	**0.364**	**0.451**	**0.445**	0.395	0.412	1.145	0.697	1.268	0.741	1.529	0.790
ETTm2	96	**0.150**	**0.241**	**0.204**	**0.302**	0.188	0.273	0.435	0.375	0.245	0.316	0.359	0.333
	192	**0.227**	**0.294**	0.282	0.358	**0.274**	**0.338**	0.995	0.494	0.413	0.401	0.756	0.396
	336	**0.254**	**0.218**	0.378	**0.425**	**0.384**	0.406	2.324	0.648	0.790	0.528	1.747	0.467
	720	**0.346**	**0.376**	0.555	**0.523**	**0.501**	0.488	9.064	1.020	2.451	0.847	6.856	0.639
	Avg	**0.244**	**0.307**	**0.355**	**0.402**	0.337	0.376	3.205	0.634	0.975	0.523	2.430	0.459
Weather	96	**0.141**	**0.203**	**0.185**	**0.244**	0.160	0.222	0.255	0.273	0.279	0.266	0.331	0.277
	192	**0.178**	**0.230**	**0.225**	**0.282**	0.202	0.265	0.390	0.353	0.337	0.316	0.498	0.345
	336	**0.241**	**0.281**	**0.274**	**0.323**	0.253	0.303	0.775	0.457	0.472	0.385	0.898	0.423
	720	**0.300**	**0.312**	0.340	0.373	**0.323**	**0.354**	2.898	0.707	0.818	0.526	2.820	0.580
	Avg	**0.215**	**0.256**	**0.256**	**0.306**	0.235	0.286	1.080	0.448	0.477	0.373	1.137	0.406
ILI	24	**1.231**	**0.771**	6.809	1.870	2.675	1.080	**4.909**	**1.329**	5.991	1.510	4.869	1.315
	36	**1.321**	**0.798**	6.850	1.890	3.081	1.194	**5.079**	**1.440**	5.922	1.539	4.917	1.422
	48	**1.412**	**0.787**	6.788	1.876	2.973	1.176	4.276	**1.339**	**4.637**	1.329	3.966	1.301
	60	**1.428**	0.808	6.908	1.893	**3.259**	**1.232**	3.855	**1.276**	4.378	1.345	3.540	1.229
	Avg	**1.348**	**0.791**	6.839	1.882	2.997	1.171	**4.530**	**1.346**	5.232	1.431	4.323	1.317

Table 11: Long-term forecasting results on additional baselines. **Bold**: the best, Underline: the second best, **Blue**: the best in deep time series model baseline.

Intervals	Methods	FFTS (FT)	Time-LLM	GPT4TS	TimesNet	PatchTST	N-HiTS	N-BEATS	ETSformer	LightTS	Dlinear	FEDformer	Stationary	Autoformer	Informer	Reformer
Yearly	SMAPE	**13.211**	13.419	15.11	15.378	13.477	13.422	13.487	18.009	14.247	16.965	14.021	13.717	13.974	14.727	16.169
	MASE	**2.949**	3.005	3.565	3.554	3.019	3.056	3.036	4.487	3.109	4.283	3.036	3.078	3.134	3.418	3.800
	OWA	**0.767**	0.789	0.911	0.918	0.792	0.795	0.795	1.115	0.827	1.058	0.811	0.807	0.822	0.881	0.973
Quarterly	SMAPE	**9.910**	10.110	10.597	10.465	10.38	10.185	10.564	13.376	11.364	12.145	11.1	10.958	11.338	11.360	13.313
	MASE	**1.120**	1.178	1.253	1.227	1.233	1.18	1.252	1.906	1.328	1.520	1.35	1.325	1.365	1.401	1.775
	OWA	**0.881**	0.889	0.938	0.923	0.921	0.893	0.936	1.302	1.000	1.106	0.996	0.981	1.012	1.027	1.252
Monthly	SMAPE	**12.542**	12.980	13.258	13.513	12.959	13.059	13.089	14.588	14.014	13.514	14.403	13.917	13.958	14.062	20.128
	MASE	**0.943**	0.963	1.003	1.039	0.970	1.013	0.996	1.368	1.053	1.037	1.147	1.097	1.103	1.141	2.614
	OWA	**0.898**	0.903	0.931	0.957	0.905	0.929	0.922	1.149	0.981	0.956	1.038	0.998	1.002	1.024	1.927
Others	SMAPE	**4.552**	4.795	6.124	6.913	4.952	4.711	6.599	7.267	15.880	6.709	7.148	6.302	5.485	24.460	32.491
	MASE	**3.112**	3.178	4.116	4.507	3.347	3.054	4.43	5.240	11.434	4.953	4.041	4.064	3.865	20.960	33.355
	OWA	0.991	1.006	1.259	1.438	1.049	**0.977**	1.393	1.591	3.474	1.487	1.389	1.304	1.187	5.879	8.679
Average	SMAPE	**11.664**	11.983	12.69	12.88	12.059	12.035	12.25	14.718	13.525	13.639	13.16	12.780	12.909	14.086	18.200
	MASE	**1.556**	1.595	1.808	1.836	1.623	1.625	1.698	2.408	2.111	2.095	1.775	1.756	1.771	2.718	4.223
	OWA	0.868	**0.859**	0.94	0.955	0.869	0.869	0.896	1.172	1.051	1.051	0.949	0.930	0.939	1.230	1.775

Table 12: Short-term forecasting results. **Bold**: the best, Underline: the second best.

Setting		FFTS		FL Baseline			Deep Time Series Model Baseline		
Intervals	Methods	FFTS (FT)	FFTS-Cen (FT)	FedAvg (FT)	FedProx (FT)	pFedMe (FT)	Time-LLM	PatchTST	N-HiTS
Yearly	SMAPE	**13.211**	13.247	13.998	14.989	14.978	13.419	13.477	13.422
	MASE	**2.999**	2.898	3.048	3.427	3.296	3.005	3.019	3.056
	OWA	**0.787**	0.792	0.797	0.888	0.890	0.789	0.792	0.795
Quarterly	SMAPE	**9.969**	9.982	10.489	10.474	10.375	10.110	10.38	10.185
	MASE	**1.120**	1.200	1.149	1.244	1.198	1.178	1.233	1.18
	OWA	**0.881**	0.892	0.900	0.912	0.907	0.889	0.921	0.893
Monthly	SMAPE	13.042	13.047	13.187	13.271	13.289	12.980	**12.959**	13.059
	MASE	**0.943**	0.955	1.042	1.102	1.090	0.963	0.970	1.013
	OWA	**0.898**	0.901	0.919	1.147	0.982	0.903	0.905	0.929
Others	SMAPE	**4.552**	4.620	4.721	4.990	4.981	4.795	4.952	4.711
	MASE	**3.112**	3.118	3.145	3.337	3.300	3.178	3.347	3.054
	OWA	**0.991**	1.013	1.101	1.123	1.124	1.006	1.049	0.977
Average	SMAPE	**11.664**	11.937	12.302	12.582	12.563	11.983	12.059	12.035
	MASE	**1.556**	1.569	1.634	1.783	1.733	1.595	1.623	1.625
	OWA	0.868	0.880	0.895	1.029	0.950	**0.859**	0.869	0.869

Table 13: Short-term forecasting results based on FL. **Bold**: the best, Underline: the second best.

Dataset	Method	FFTS		FFTS-Cen		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	12.5%	**0.014**	0.076	0.015	**0.078**	**0.017**	**0.085**	0.023	0.101	0.041	0.130	0.096	0.229	0.093	0.206	0.080	0.193	0.052	0.166	0.032	0.119	0.046	0.144	0.063	0.180	0.042	0.146
	25%	**0.014**	**0.078**	0.015	0.084	**0.022**	**0.096**	0.023	0.101	0.044	0.135	0.096	0.229	0.093	0.206	0.080	0.193	0.052	0.166	0.032	0.119	0.046	0.144	0.063	0.180	0.042	0.146
	37.5%	0.034	**0.098**	0.036	0.097	**0.029**	**0.111**	**0.029**	**0.111**	0.049	0.143	0.133	0.271	0.113	0.231	0.103	0.219	0.069	0.191	**0.039**	0.131	0.057	0.161	0.079	0.200	0.063	0.182
	50%	0.031	**0.111**	**0.030**	0.114	**0.040**	0.128	0.036	**0.124**	0.055	0.151	0.186	0.323	0.134	0.255	0.132	0.248	0.089	0.218	0.047	0.145	0.067	0.174	0.093	0.218	0.082	0.208
	Avg	**0.023**	**0.091**	0.024	0.093	**0.028**	**0.105**	0.027	0.107	0.047	0.140	0.120	0.253	0.104	0.218	0.093	0.206	0.062	0.177	0.036	0.126	0.051	0.150	0.071	0.188	0.055	0.166
ETTh2	12.5%	**0.014**	**0.067**	0.015	0.070	**0.017**	**0.076**	0.018	0.080	0.026	0.094	0.108	0.239	0.034	0.127	0.062	0.166	0.056	0.159	0.021	0.088	0.023	0.092	0.133	0.270	0.108	0.228
	25%	**0.017**	**0.071**	0.018	0.074	**0.020**	**0.080**	0.020	0.085	0.028	0.099	0.164	0.294	0.042	0.143	0.085	0.196	0.080	0.195	0.024	0.096	0.026	0.101	0.135	0.272	0.136	0.262
	37.5%	**0.018**	**0.072**	**0.018**	0.078	**0.022**	**0.087**	0.023	0.091	0.030	0.104	0.237	0.356	0.051	0.159	0.106	0.222	0.110	0.231	0.027	0.103	0.030	0.108	0.155	0.293	0.175	0.300
	50%	0.022	**0.082**	**0.020**	0.086	**0.025**	**0.095**	0.026	0.098	0.034	0.110	0.323	0.421	0.059	0.174	0.131	0.247	0.156	0.276	0.030	0.108	0.035	0.119	0.200	0.333	0.211	0.329
	Avg	**0.018**	**0.073**	**0.018**	0.077	0.021	**0.084**	**0.022**	0.088	0.029	0.102	0.208	0.327	0.046	0.151	0.096	0.208	0.101	0.215	0.026	0.099	0.029	0.105	0.156	0.292	0.157	0.280
ETTm1	12.5%	**0.032**	**0.121**	0.033	0.125	**0.043**	**0.140**	0.057	0.159	0.093	0.201	0.126	0.263	0.240	0.345	0.151	0.267	0.070	0.190	0.060	0.165	0.074	0.182	0.114	0.234	0.074	0.194
	25%	0.042	**0.137**	**0.040**	0.140	**0.054**	**0.156**	0.069	0.178	0.107	0.217	0.169	0.304	0.265	0.364	0.180	0.292	0.106	0.236	0.080	0.189	0.090	0.203	0.140	0.262	0.102	0.227
	37.5%	0.065	0.156	**0.064**	**0.153**	**0.072**	**0.180**	0.084	0.196	0.120	0.230	0.220	0.347	0.296	0.382	0.215	0.318	0.124	0.258	0.102	0.212	0.109	0.222	0.174	0.293	0.135	0.261
	50%	**0.092**	0.198	0.090	**0.195**	**0.107**	0.216	0.102	**0.215**	0.141	0.248	0.293	0.402	0.334	0.404	0.257	0.347	0.165	0.299	0.133	0.240	0.137	0.248	0.215	0.325	0.179	0.298
	Avg	0.058	**0.153**	**0.057**	**0.153**	**0.069**	**0.173**	0.078	0.187	0.115	0.224	0.202	0.329	0.284	0.373	0.201	0.306	0.117	0.246	0.094	0.201	0.103	0.214	0.161	0.279	0.122	0.245
ETTm2	12.5%	0.029	0.101	**0.027**	**0.099**	**0.039**	**0.125**	0.040	0.130	0.057	0.152	0.187	0.319	0.101	0.231	0.100	0.216	0.095	0.212	0.042	0.133	0.044	0.138	0.305	0.431	0.163	0.289
	25%	**0.032**	**0.132**	0.033	**0.130**	**0.044**	0.135	0.046	0.141	0.061	0.158	0.279	0.390	0.115	0.246	0.127	0.247	0.137	0.258	0.049	0.147	0.050	0.149	0.322	0.444	0.206	0.331
	37.5%	0.058	**0.155**	**0.062**	**0.150**	**0.051**	**0.147**	0.052	0.151	0.067	0.166	0.400	0.465	0.126	0.257	0.158	0.276	0.187	0.304	0.056	0.158	0.060	0.163	0.353	0.462	0.252	0.370
	50%	**0.058**	**0.147**	**0.058**	0.148	0.059	**0.158**	**0.060**	0.162	0.073	0.174	0.602	0.572	0.136	0.268	0.183	0.299	0.232	0.341	0.065	0.170	0.068	0.173	0.369	0.472	0.316	0.419
	Avg	**0.044**	**0.134**	0.045	0.132	**0.048**	**0.141**	0.049	0.146	0.065	0.163	0.367	0.436	0.119	0.250	0.142	0.259	0.163	0.279	0.053	0.152	0.055	0.156	0.337	0.452	0.234	0.352
Weather	12.5%	0.021	0.048	**0.020**	0.052	0.026	0.049	0.025	**0.045**	0.029	0.049	0.057	0.141	0.047	0.101	0.039	0.084	0.041	0.107	0.027	0.051	**0.026**	**0.047**	0.037	0.093	0.031	0.076
	25%	**0.026**	0.057	**0.024**	0.056	0.028	**0.052**	0.029	**0.052**	0.031	0.053	0.065	0.155	0.052	0.111	0.048	0.103	0.064	0.163	0.029	0.056	0.030	**0.054**	0.042	0.100	0.035	0.082
	37.5%	0.033	0.076	0.034	0.062	0.033	**0.060**	**0.031**	0.057	0.035	**0.058**	0.081	0.180	0.058	0.121	0.057	0.117	0.107	0.229	0.033	0.062	**0.032**	**0.060**	0.049	0.111	0.040	0.091
	50%	0.035	**0.058**	**0.038**	0.060	**0.037**	0.065	**0.034**	**0.062**	**0.038**	**0.063**	0.102	0.207	0.065	0.133	0.066	0.134	0.183	0.312	**0.037**	0.068	**0.037**	0.067	0.053	0.114	0.046	0.099
	Avg	**0.029**	0.060	**0.029**	0.058	0.031	**0.056**	0.030	**0.054**	0.060	0.144	0.076	0.171	0.055	0.117	0.052	0.110	0.099	0.203	0.032	0.059	0.031	0.057	0.045	0.104	0.038	0.087

Table 14: Full results on imputation task. **Bold**: the best, Underline: the second best, **Blue**: the best in deep time series model baseline.

Method	SMD			MSL			SMAP			SWaT			PSM			Avg F1
Metrics	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	%
Transformer	83.58	76.13	79.56	71.57	87.37	78.68	89.37	57.12	69.70	68.84	96.53	80.37	62.75	96.56	76.07	76.88
LogTransformer	83.46	70.13	76.21	73.05	87.37	79.57	89.15	57.59	69.97	68.67	97.32	80.52	63.06	98.00	76.74	76.60
Autoformer	88.06	82.35	85.11	77.27	80.92	79.05	90.40	58.62	71.12	89.85	95.81	92.74	99.08	88.15	93.29	84.26
Pyraformer	85.61	80.61	83.04	83.81	85.93	84.86	92.54	57.71	71.09	87.92	96.00	91.78	71.67	96.02	82.08	82.57
Informer	86.60	77.23	81.65	81.77	86.48	84.06	90.11	57.13	69.92	70.29	96.75	81.43	64.27	96.33	77.10	78.83
Reformer	82.58	69.24	75.32	85.51	83.31	84.40	90.91	57.44	70.40	72.50	96.53	82.80	59.93	95.38	73.61	77.31
ETSformer	87.44	79.23	83.13	85.13	84.93	**85.03**	92.25	55.75	69.50	90.02	80.36	84.91	99.31	85.28	91.76	82.87
FEDformer	87.95	82.39	85.08	77.14	80.07	78.57	90.47	58.10	70.76	90.17	96.42	93.19	97.31	97.16	97.23	84.97
Stationary	88.33	81.21	84.62	68.55	89.14	77.50	89.37	59.02	71.09	68.03	96.75	79.88	97.82	96.76	97.29	82.08
LightTS	87.10	78.42	82.53	82.40	75.78	78.95	92.58	55.27	69.21	91.98	94.72	93.33	98.37	95.97	97.15	84.23
Anomaly Transformer	88.91	82.23	85.49	79.61	87.37	83.31	91.85	58.11	71.18	72.51	97.32	83.10	68.35	94.72	79.40	80.50
TimesNet	87.91	81.54	84.61	89.54	75.36	81.84	90.14	56.40	69.39	90.75	95.40	93.02	98.51	96.20	97.34	85.24
PatchTST	87.26	82.14	84.62	88.34	70.96	78.70	90.64	55.46	68.82	91.10	80.94	85.72	98.84	93.47	96.08	82.79
DLinear	83.62	71.52	77.10	84.34	85.42	84.88	92.32	55.41	69.26	80.91	95.30	87.52	98.28	89.26	93.55	82.46
GPT4TS	88.89	84.98	86.89	82.00	82.91	82.45	90.60	60.95	72.88	92.20	96.34	94.23	98.62	95.68	97.13	86.72
FedAvg	88.50	85.96	87.21	85.00	81.71	83.33	76.50	72.03	74.20	94.02	96.00	95.00	98.50	97.90	98.20	87.59
FedProx	87.20	83.78	85.45	80.49	84.00	82.21	67.00	73.50	70.09	91.65	94.00	92.81	97.20	95.99	96.59	85.43
pFedMe	87.50	84.13	85.78	81.17	85.00	83.04	74.00	68.45	71.11	90.53	93.50	91.99	97.00	95.41	96.20	85.62
FFTS-Cen (FT)	89.30	86.60	87.93	85.50	81.70	83.56	76.50	71.80	74.08	95.70	94.20	94.94	98.20	97.70	97.95	87.69
FFTS (Ours, FT)	89.80	87.08	**88.42**	86.00	82.12	84.01	77.00	72.21	**74.53**	96.00	94.55	**95.27**	98.50	98.00	**98.25**	**88.10**

Table 15: Full results of anomaly detection with deep time series model. **Bold**: the best, Underline: the second best.

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